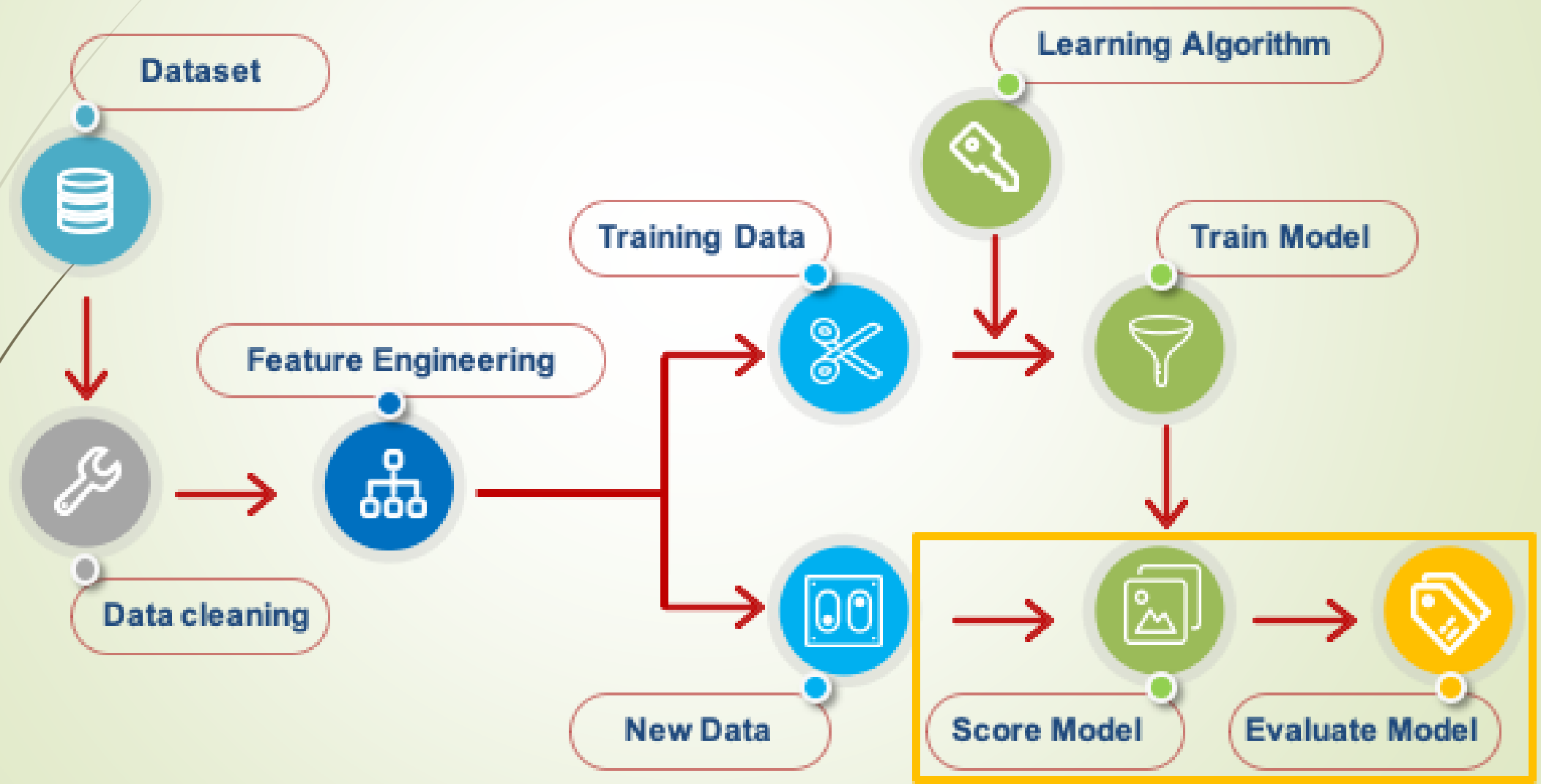




Metrik Evaluasi Kinerja Machine Learning

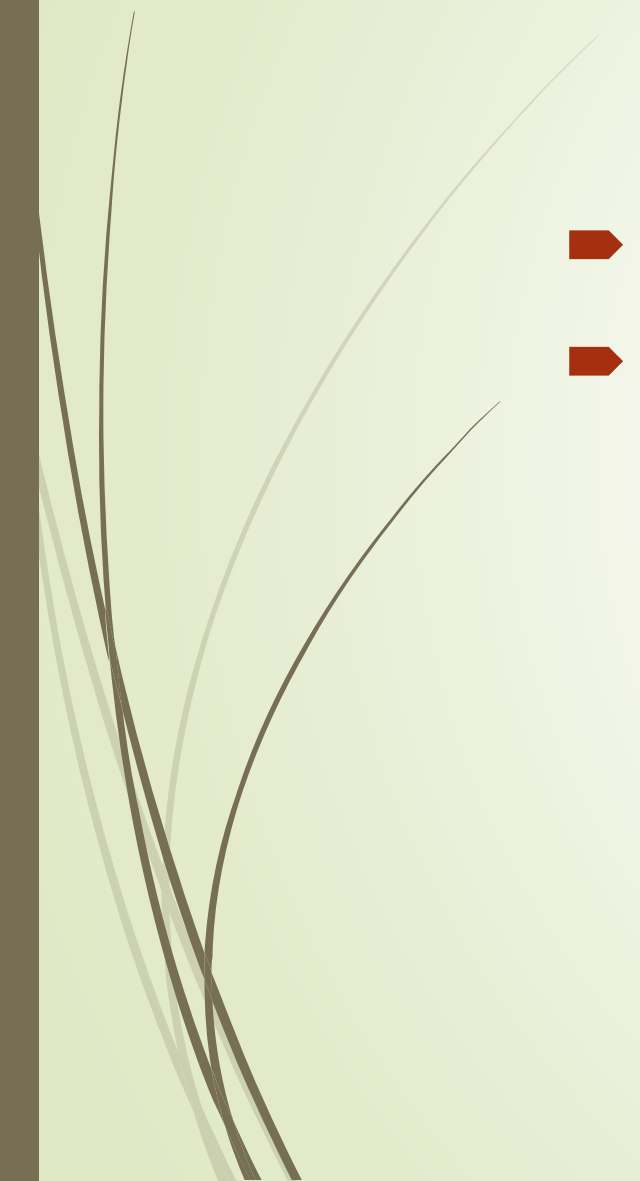
Dr. Jasman Pardede, S.Si., M.T

Tahapan pembelajaran mesin





Evaluasi Kinerja

- Permasalahan Regresi
 - Permasalahan Klasifikasi
- 

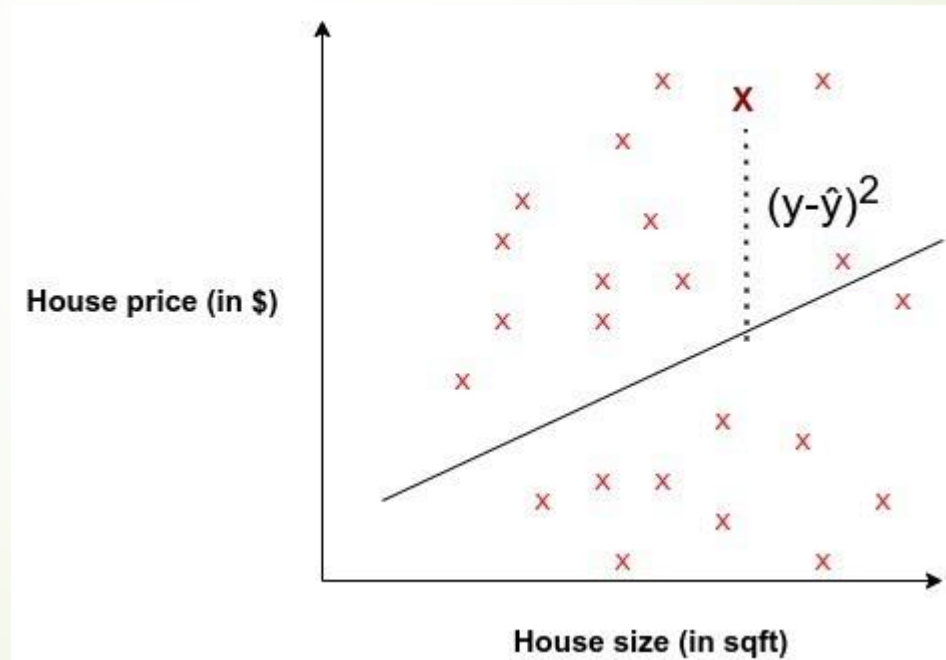
Kinerja Regresi

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)
- R^2 Coefficient of determination



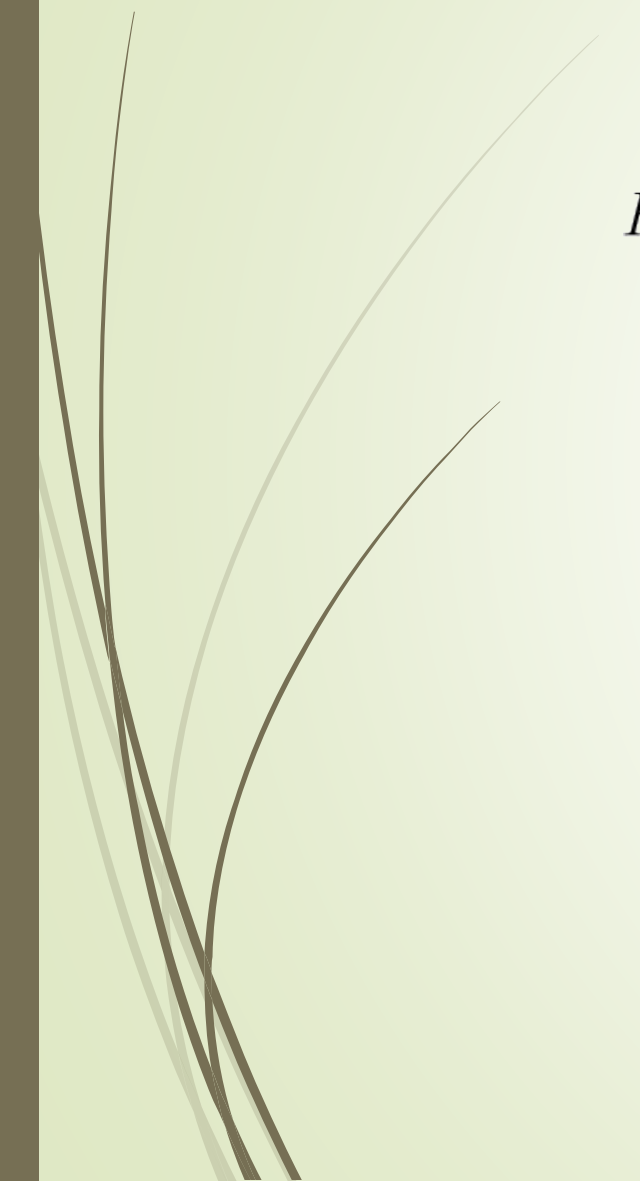
Mean Squared Error

$$MSE = \frac{1}{N} \sum_{j=1}^N (y_j - \hat{y}_j)^2$$



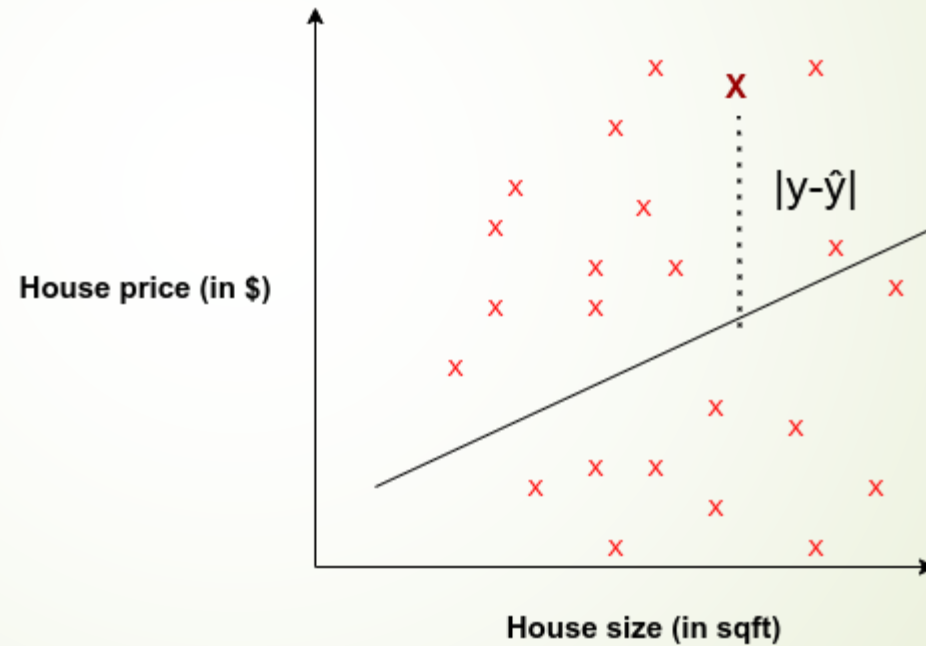


Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (y_j - \check{y}_j)^2}$$


Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{j=1}^N |y_j - \tilde{y}_j|$$





Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{\sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right|}{n} \quad (6.9)$$

R² Coefficient of determination

```
SE_line = sum((y-y_hat)**2)
SE_mean = sum((y-y.mean())**2)
r2 = 1-(SE_line/SE_mean)
```

$$R^2 = 1 - \frac{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2}$$

Contoh kasus

Tabel 6. 2. Contoh data status aktual dan hasil prediksi model regresi

Pengamatan	Aktual (A)	Prediksi (F)
1	108	111.20
2	129	124.43
3	126	133.24
4	149	146.47
5	168	155.29
6	161	168.51
7	174	175.86

Tabel 6. 3. Data pembantu perhitungan kinerja model regresi

#	Aktual (A)	Prediksi (F)	A-F	(A-F)	(A-F) ²	(A-F)/A
1	108	111.20	3.203776	-3.20	10.26418	0.0296
2	129	124.43	4.571615	4.57	20.89966	0.0354
3	126	133.24	7.244792	-7.24	52.48701	0.0575
4	149	146.47	2.530599	2.53	6.403931	0.0170
5	168	155.29	12.71419	12.71	161.6507	0.0757
6	161	168.51	7.510417	-7.51	56.40636	0.0466
7	174	175.86	1.857422	-1.86	3.450016	0.0107
Jumlah	1015	1015	39.63281	3.98E-12	311.5618	0.2725

$$MAE = \frac{\sum_{i=1}^n |A_i - F_i|}{n} = \frac{39.63281}{7} = 5.66.$$

$$MAPE = \frac{\sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right|}{n} = \frac{0.2725}{7} = 0.0389.$$

$$MSE = \frac{\sum_{i=1}^n (A_i - F_i)^2}{n} = \frac{311.5618}{7} = 44.48.$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (A_i - F_i)^2}{n}} = \sqrt{44.48} = 6.67.$$

Tabel 6. 4. Data pembantu perhitungan kinerja R^2

#	A	F	A ²	F ²	AF
1	108	111.20	11664	12366.28	12010.01
2	129	124.43	16641	15482.42	16051.26
3	126	133.24	15876	17754.17	16788.84
4	149	146.47	22201	21453.29	21823.94
5	168	155.29	28224	24113.68	26088.02
6	161	168.51	25921	28395.76	27130.18
7	174	175.86	30276	30925.83	30599.19
Jumlah	1015	1015	150803	150491.4	150491.4

$$R^2 = \frac{n \sum A_i F_i - \sum A_i \sum F_i}{\sqrt{(n \sum A_i^2 - (\sum A_i)^2)(n \sum F_i^2 - (\sum F_i)^2)}} = \frac{7 * 150491.4 - (1015)(1015)}{\sqrt{(7 * (150803) - (1015)^2)(7 * (150491.4) - (1015)^2)}}$$

$$R^2 = \frac{23215.1}{24281.1} = 0.9561.$$

```

1 import math
2
3 actual = [108, 129, 126, 149, 168, 161, 174]
4 predicted = [111.20, 124.43, 133.24, 146.47, 155.29, 168.51, 175.86]
5
6 n = len(actual)
7 print("Banyaknya data: ", n)
8 def fungsi_mae(actual, predicted):
9     sum = 0
10    for i in range(n):
11        sum += abs(actual[i] - predicted[i])
12    return sum/n
13
14 def fungsi_mse(actual, predicted):
15    sum = 0
16    for i in range(n):
17        sum += pow((actual[i] - predicted[i]), 2)
18    return sum/n
19
20 def fungsi_rmse(actual, predicted):
21    sum = 0
22    for i in range(n):
23        sum += pow((actual[i] - predicted[i]), 2)
24    return math.sqrt(sum/n)
25
26 def fungsi_mape(actual, predicted):
27    sum = 0
28    for i in range(n):
29        sum += abs((actual[i] - predicted[i])/actual[i])
30    return sum/n
31
32 print("Nilai MAE: ", fungsi_mae(actual=actual, predicted=predicted))
33 print("Nilai MSE: ", fungsi_mse(actual=actual, predicted=predicted))
34 print("Nilai RMSE: ", fungsi_rmse(actual=actual, predicted=predicted))
35 print("Nilai MAPE: ", fungsi_mape(actual=actual, predicted=predicted))

```

```

Banyaknya data: 7
Nilai MAE: 5.6600000000000003
Nilai MSE: 44.47817142857145
Nilai RMSE: 6.669195710771386
Nilai MAPE: 0.03892664989442879

```

```
1 import math
2 from sklearn.metrics import mean_absolute_error as mae
3 from sklearn.metrics import mean_squared_error as mse
4 from sklearn.metrics import mean_absolute_percentage_error as mape
5 from sklearn.metrics import r2_score as r2
6
7 actual = [108, 129, 126, 149, 168, 161, 174]
8 predicted = [111.20, 124.43, 133.24, 146.47, 155.29, 168.51, 175.86]
9
10 print("Nilai MAE: ", mae(actual, predicted))
11 print("Nilai MSE: ", mse(actual, predicted))
12 print("Nilai RMSE: ", math.sqrt(mse(actual, predicted)))
13 print("Nilai MAPE: ", mape(actual, predicted))
14 print("Nilai r Square: ", r2(actual, predicted))
```

```
Nilai MAE: 5.6600000000000003
Nilai MSE: 44.47817142857145
Nilai RMSE: 6.669195710771386
Nilai MAPE: 0.03892664989442879
Nilai r Square: 0.9141821389195148
```

Contoh Kasus 2

No	Data Aktual	Data Prediksi	Mean Square Error	Root Mean Squared Error	Mean Absolute Error	SE_Line	SE_Mean	R ² Coefficient of determination
1	80	80.1	0.01	0.01	0.1	0.01	6.76	
2	84	83.5	0.25	0.25	0.5	0.25	1.96	
3	79	78.5	0.25	0.25	0.5	0.25	12.96	
4	81	80.4	0.36	0.36	0.6	0.36	2.56	
5	89	87.5	2.25	2.25	1.5	2.25	40.96	
	82.6		0.624	0.624	0.64	0.624	13.04	

0.789937

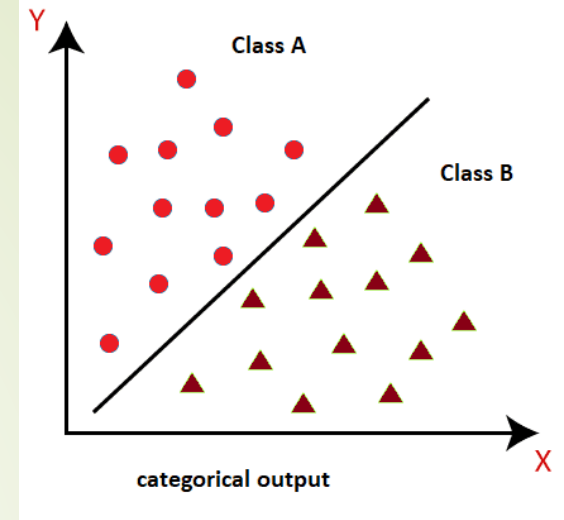
0.04785276

0.952147239

SE_Line/SE_Mean

Kinerja Klasifikasi

- Accuracy
- Confusion Matrix (not a metric but fundamental to others)
- Precision and Recall
- F1-score
- Specificity
- AU-ROC



Confusion Matrix

- **True Positive (TP)** : kasus dimana mahasiswa diprediksi (Positif) DO, memang benar(True) DO.
- **True Negative (TN)** : kasus dimana mahasiswa diprediksi tidak(Negatif) DO dan sebenarnya mahasiswa tersebut memang (True) tidak DO.
- **False Positive (FP)** : kasus dimana mahasiswa yang diprediksi positif DO, ternyata tidak DO. Prediksinya salah (False).
- **False Negatif (FN)**: kasus dimana mahasiswa yang diprediksi tidak DO (Negatif), tetapi ternyata sebenarnya(TRUE) DO.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN



Accuracy

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

Precision and Recall

$$\textit{Precision} = \frac{TP}{TP + FP}$$

$$\textit{Recall} = \frac{TP}{TP + FN}$$

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN



F1-score

$$F1 = 2 * (precision * recall) / (precision + recall)$$

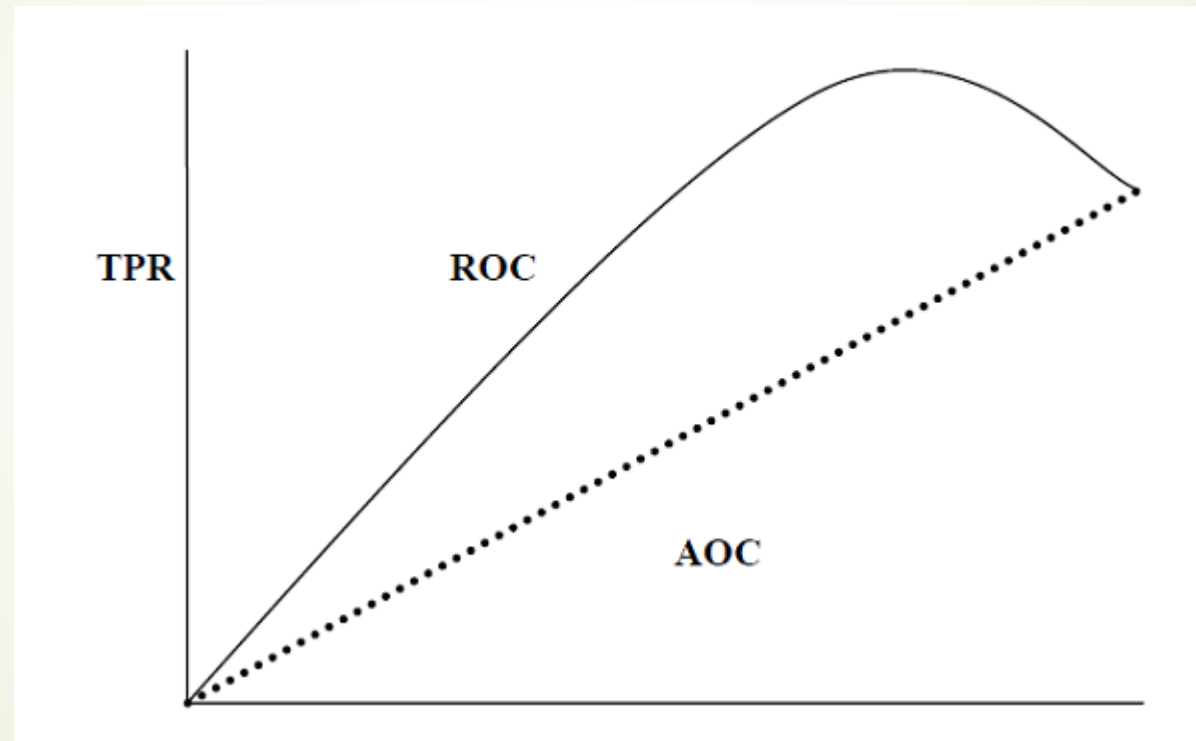
Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

AU-ROC (Area Under ROC curve)

- Metrik AUC-ROC akan memberi tahu kita tentang kemampuan model dalam membedakan kelas



$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

Contoh Kasus

Nim	Status Sebenarnya	Hasil Prediksi
001	Tidak DO	Tidak DO
002	Tidak DO	Tidak DO
003	Tidak DO	Tidak DO
004	Tidak DO	DO
005	Tidak DO	DO
006	DO	Tidak DO
007	DO	DO
008	DO	DO
009	DO	DO
010	DO	DO

TN

FP

FN

TP

Prediksi DO

Prediksi			
		DO	Tidak DO
TP 4	FP 2	Positif	
FN 1	TN 3	Negatif	
		DO	Tidak DO
		Positif	Negatif
		Aktual	

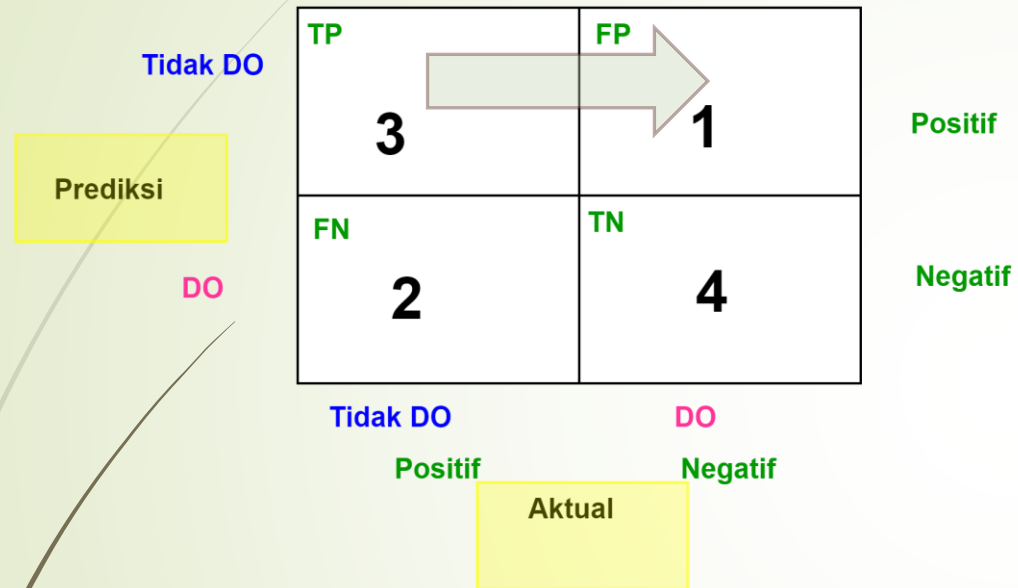
		Prediksi	
		DO	Tidak DO
TP 4	FN 1	Positif	
FP 2	TN 3	Negatif	
		DO	Tidak DO
		Positif	Negatif
Aktual			

$$Precision = \frac{TP}{TP + FP}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

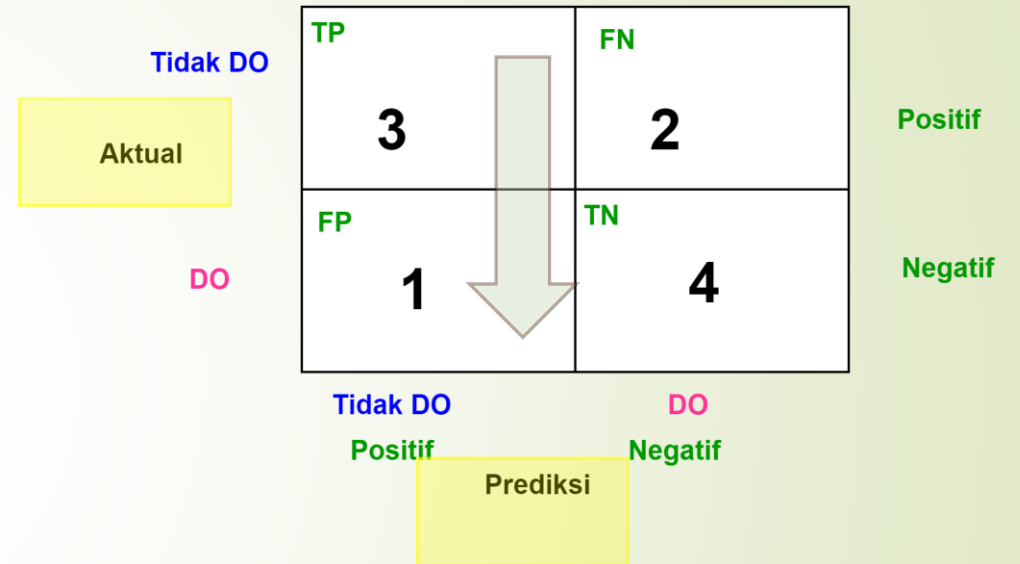
Prediksi **Tidak DO**



$$Precision = \frac{TP}{TP + FP}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Recall = \frac{TP}{TP + FN}$$



Evaluasi Kinerja

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$F1 = 2 * (precision * recall) / (precision + recall)$$

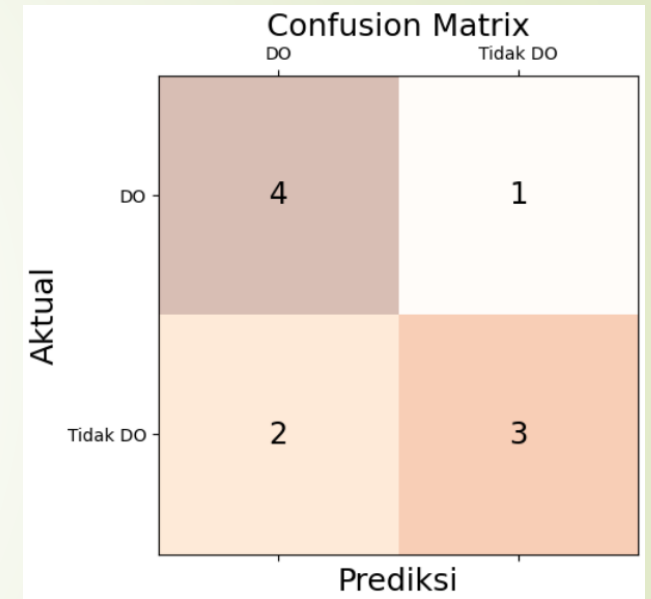
Evaluasi Kinerja (2)

- Akurasi = $(4+3) / (4+2+1+3) = 7/10 = 70\%$
- Precision = $4 / (4+2) = 4/6 = 67\%$
- Recall = $4/(4+1) = 4/5 = 80\%$.
- Specificity = $(3) / (3+2) = 3/5 = 60\%$
- F1 Score = $2 * (80\% * 67\%) / (80\% + 67\%) = 72,93\%$

```

1 from sklearn.metrics import confusion_matrix
2 from sklearn.metrics import precision_score, recall_score, f1_score,
  accuracy_score
3 import matplotlib.pyplot as plt
4
5 y_true = ['DO', 'DO', 'DO', 'DO', 'DO', 'Tidak DO', 'Tidak DO', 'Tidak DO', 'Tidak
  DO', 'Tidak DO']
6 y_pred = ['DO', 'DO', 'DO', 'DO', 'Tidak DO', 'DO', 'DO', 'Tidak DO', 'Tidak DO',
  'Tidak DO']
7 labels = ['DO', 'Tidak DO']
8
9 conf_matrix = confusion_matrix(y_true=y_true, y_pred=y_pred, labels=labels)
10 fig, ax = plt.subplots(figsize=(5, 5))
11 ax.matshow(conf_matrix, cmap=plt.cm.Oranges, alpha=0.3)
12 for i in range(conf_matrix.shape[0]):
13     for j in range(conf_matrix.shape[1]):
14         ax.text(x=j, y=i, s=conf_matrix[i, j], va='center', ha='center', size='xx-
  large')
15 ax.set_xticklabels([""] + labels)
16 ax.set_yticklabels([""] + labels)
17 plt.xlabel('Prediksi', fontsize=18)
18 plt.ylabel('Aktual', fontsize=18)
19 plt.title('Confusion Matrix', fontsize=18)
20 plt.show()
21
22 print('Accuracy: %.3f % accuracy_score(y_true, y_pred))
23 print('Precision: %.3f % precision_score(y_true, y_pred, pos_label='DO'))
24 print('Recall: %.3f % recall_score(y_true, y_pred, pos_label='DO'))
25 print('F1 Score: %.3f % f1_score(y_true, y_pred, pos_label='DO'))

```

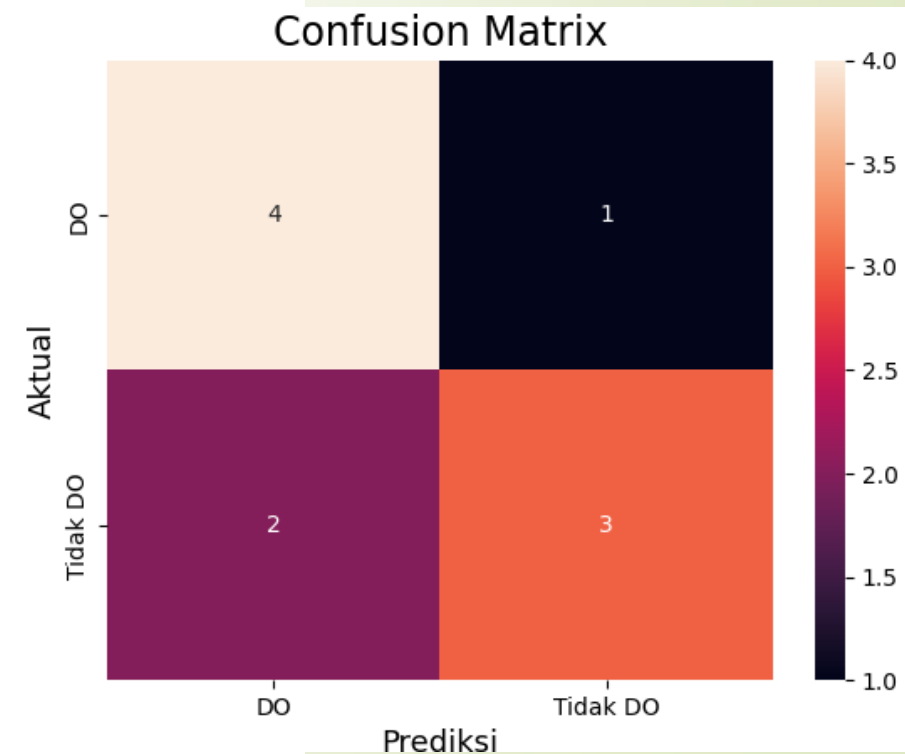


```

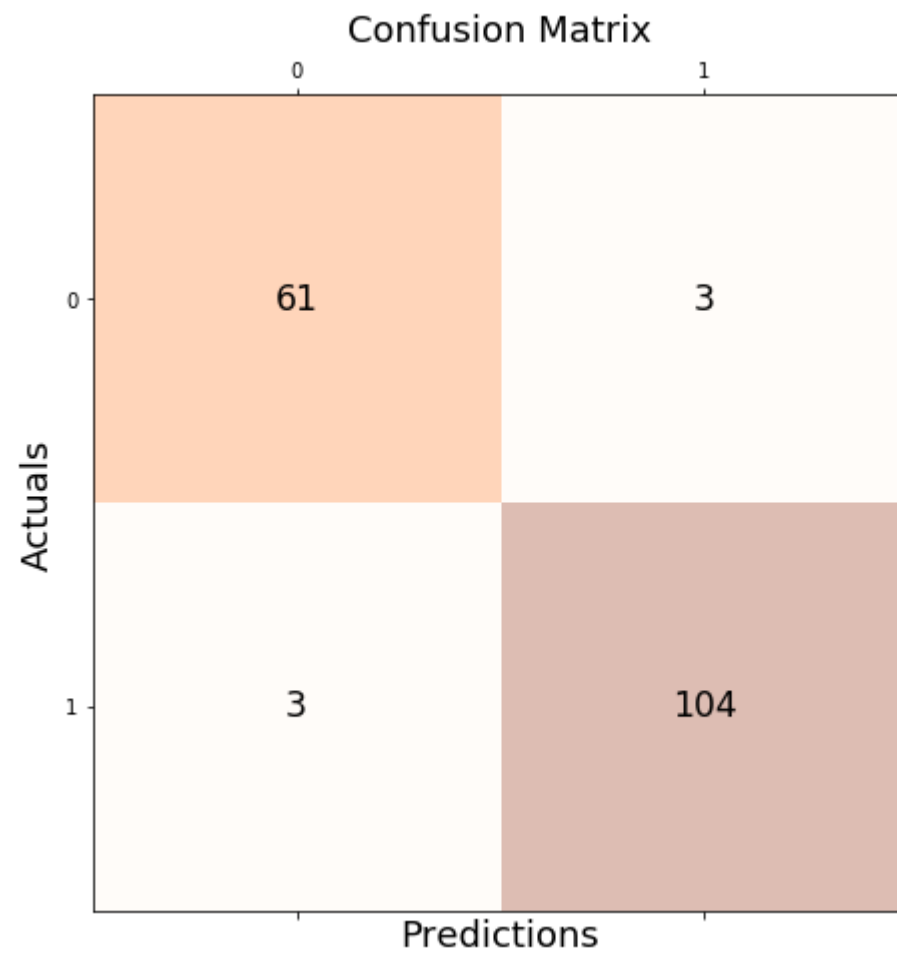
1 import numpy as np
2 from sklearn.metrics import confusion_matrix, classification_report
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import pandas as pd
6
7 actual = np.array(['DO','DO','DO','DO','DO','Tidak DO','Tidak DO','Tidak DO','Tidak DO','Tidak DO'])
8 predicted = np.array(['DO','DO','DO','DO', 'Tidak DO','DO','DO','Tidak DO','Tidak DO','Tidak DO'])
9
10 df_act = pd.DataFrame(actual)
11 df_act.columns = ["Aktual"]
12 df_pre = pd.DataFrame(predicted)
13 df_pre.columns = ["Prediksi"]
14 df = pd.concat([df_act, df_pre], axis='columns')
15
16 print(df)
17 cm = confusion_matrix(actual,predicted)
18 sns.heatmap(cm,
19             annot=True,
20             fmt='g',
21             xticklabels=['DO','Tidak DO'],
22             yticklabels=['DO','Tidak DO'])
23
24 plt.ylabel('Prediksi',fontsize=13)
25 plt.xlabel('Aktual',fontsize=13)
26 plt.title('Confusion Matrix',fontsize=17)
27 plt.show()
28
29 print(classification_report(actual, predicted))

```

	Aktual	Prediksi				
0	DO	DO				
1	DO	DO				
2	DO	DO				
3	DO	DO				
4	DO	Tidak DO				
5	Tidak DO	DO				
6	Tidak DO	DO				
7	Tidak DO	Tidak DO				
8	Tidak DO	Tidak DO				
9	Tidak DO	Tidak DO				
			precision	recall	f1-score	support
	DO		0.67	0.80	0.73	5
	Tidak DO		0.75	0.60	0.67	5
accuracy					0.70	10
macro avg			0.71	0.70	0.70	10
weighted avg			0.71	0.70	0.70	10



Test





Evaluasi Kinerja Klustering

- Silhouette score
- Rand Indeks
- Adjusted Rand Indeks
- Mutual Information
- Calinski-Harabasz Indeks
- Davies-Bouldin Index



Evaluasi Kinerja Asosiasi

- ▀ Key performance indicators (KPI)



Ciri-ciri KPI

- Relevan: harus relevan dengan tujuan dan strategi bisnis organisasi
- Jelas: harus dipahami dengan jelas oleh semua pihak
- Dapat Diukur: menggunakan angka atau data yang dapat diukur
- Dapat Dicapai: harus realistis dan dapat dicapai dalam cangkang waktu tertentu
- Berprogres: dapat mengukur kemajuan perusahaan dalam mencapai tujuan
- Fleksibel: harus fleksibel dan dapat diperbaharui sesuai perubahan bisnis